

LAB - 03 ARCHITECTURAL JUSTIFICATION

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PROBLEM STATEMENT: #37

Group Travel Expense Simplification Engine

➤ Architecture Selection

We chose **Layered Architecture** for the Group Travel Expense Simplification Engine.

➤ Architectural Choice

We selected Layered Architecture because the system has different types of work that can be clearly separated. The **Presentation Layer** handles the interaction with travellers and the Trip Admin. The **Business Layer** manages expenses, payments, currency conversion, and debt settlement, while the **Data Layer** stores the required trip and expense information.

➤ Reason 1

One reason for choosing this architecture is that it keeps the main functions of the system organized. For example, the **Order Manager** can handle recording and managing expenses, while the **Settlement Engine** can focus on calculating traveller balances and finding the minimum set of payments needed to settle the group.

➤ Reason 2

Another reason is that the system has several related operations such as recording expenses, converting currencies, calculating balances, and making payments. Keeping these responsibilities in separate components makes the system easier to update or change without affecting the other parts of the system.

➤ Security Advantage

Layered Architecture also provides a level of security by keeping the database separate from the User Interface. Users do not directly access stored expense and payment information. Requests can be checked and validated in the Business Layer before information is stored or retrieved.

➤ Performance Benefit

The **Settlement Engine** handles the min-flow debt settlement calculation separately from the other components. This allows the settlement calculation to be optimized without affecting the rest of the system and supports the requirement that settlement calculations should complete within **100 ms for up to 50 travellers and 500 expenses**.